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Predicting the Trainability of Variational Quantum Circuits: A Data-Driven Model for Barren Plateaus

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Data-generation simulator and interactive demonstrations are available at <https://quantum.qbithabib.com>. Code, dataset-generation scripts, and random seeds are released openly (see the reproducibility statement).

ABSTRACT Barren plateaus, the exponential vanishing of cost-function gradients as a variational quantum circuit is scaled, are the central obstacle to training quantum machine-learning models, and diagnosing them today requires expensive gradient sampling on each candidate circuit. We reframe barren-plateau diagnosis as a supervised prediction problem: given only a circuit architecture, namely qubit count, depth, ansatz type, entanglement pattern, entangler gate, and cost-observable locality, can a classical model predict its trainability without sampling a single gradient? Using an exact statevector simulator we generate a dataset of 20,000 random variational circuits, each labeled by the empirical gradient variance estimated over 200 random parameter vectors via the parameter-shift rule. We compare six model families and find that gradient-boosted trees predict the base-ten logarithm of the gradient variance with a coefficient of determination of 0.685 on held-out architectures, and separate barren from trainable circuits with an area under the receiver operating characteristic curve of 0.991. Crucially, in an extrapolation setting that trains only on small circuits and tests on the two largest, unseen circuit sizes, the predictor retains a coefficient of determination of 0.719, indicating that the architectural signature of a barren plateau transfers to unseen circuit sizes. Feature-importance analysis recovers established theory: cost-observable locality and entanglement structure dominate. We make no quantum-advantage claim; this is classical machine learning about quantum circuits, offering a cheap ansatz-screening heuristic and a data-driven view of which design choices cause untrainability.

INDEX TERMS Barren plateaus, gradient boosting, gradient variance, quantum machine learning, statevector simulation, trainability, variational quantum circuits.

I. INTRODUCTION

VARIATIONAL quantum algorithms (VQAs) are the leading candidate for extracting value from noisy intermediate-scale quantum (NISQ) hardware [1]. A VQA encodes a problem into a parameterized quantum circuit (an *ansatz*), measures a cost function $C(\theta) = \langle \psi(\theta) | O | \psi(\theta) \rangle$, and updates the parameters θ with a classical optimizer. This template underlies variational quantum eigensolvers, quantum approximate optimization, and most quantum machine-learning (QML) classifiers [2]–[4].

The dominant obstacle to scaling this template is the *barren plateau* (BP) phenomenon: for broad classes of ansätze the variance of the cost gradient vanishes exponentially in the number of qubits, so the loss landscape becomes almost

everywhere flat and gradient-based training stalls [5], [6]. Barren plateaus are now known to be triggered by *global* cost observables [7], by highly expressive or deep ansätze [8], by poor initialization [9], and by hardware noise [10]. Whether a given architecture suffers from a barren plateau is therefore a property of its *design*, not merely of a particular parameter setting.

The conventional way to check whether a candidate circuit is trainable is to *measure* it: sample many random parameter vectors, evaluate gradients, and estimate their variance. This is exactly the cost we would like to avoid when screening a large design space, and it becomes infeasible precisely in the large-qubit regime where barren plateaus matter most. This motivates our central question: can a classical model predict a

variational circuit's trainability directly from its architectural description, without sampling any gradients?

We answer this empirically. We treat trainability prediction as a supervised learning problem over circuit *specs*. The features are cheap, integer-valued architectural descriptors; the label is the gradient variance, obtained once, offline, by exact statevector simulation. We then ask whether a model trained on this data (i) generalizes to unseen architectures, and (ii) *extrapolates* from small, easily simulated circuits to larger ones, the regime where a cheap predictor would be most useful.

The contributions of this article are as follows.

- We formulate barren-plateau diagnosis as a supervised prediction task, mapping a variational circuit's architecture to its gradient-variance trainability label, cast as both regression and binary classification.
- We release a reproducible pipeline that generates a labeled dataset of 20,000 random circuits using an exact, dependency-light statevector simulator, with fixed seeds and streaming output.
- We compare six model families and show gradient-boosted trees predict trainability with a coefficient of determination of 0.685 on held-out architectures and separate barren circuits with an area under the curve of 0.991.
- We demonstrate extrapolation across scale: trained only on $n \leq 10$ qubits, the model still attains a coefficient of determination of 0.719 on larger, unseen circuit sizes.
- We recover the established theory through permutation feature importance, giving the model an interpretable, physics-consistent basis.

We emphasize honest framing: we make no claim of quantum advantage. The model is classical, the data is classically simulated, and the contribution is a data-driven diagnostic for a known quantum-computing bottleneck [11], [12].

II. BACKGROUND AND RELATED WORK

A. VARIATIONAL QUANTUM CIRCUITS

A hardware-efficient ansatz [13] alternates layers of single-qubit rotations with a fixed pattern of two-qubit entangling gates. With n qubits and L layers the state is

$$|\psi(\theta)\rangle = \prod_{\ell=1}^L \left[U_{\text{ent}} \prod_{q=1}^n R_{a_{\ell,q}}(\theta_{\ell,q}) \right] |0\rangle^{\otimes n}, \quad (1)$$

where R_a is a Pauli rotation about axis $a \in \{x, y, z\}$ and U_{ent} is a layer of controlled-Z or controlled-X gates arranged linearly, circularly, or all-to-all. The cost is the expectation of a Pauli-Z string, with $O = Z_0$ (a *local* cost) or $O = Z_0 Z_1 \cdots Z_{n-1}$ (a *global* cost).

B. BARREN PLATEAUS

A circuit family exhibits a barren plateau if $\text{Var}_{\theta}[\partial C / \partial \theta_k]$ decays exponentially in n [5]. Cerezo *et al.* [7] proved that global cost functions induce barren plateaus even at shallow

depth, whereas local costs can remain trainable up to logarithmic depth, making cost locality a first-order design knob. Expressibility [8] and entanglement likewise correlate with flatter landscapes; structured architectures such as quantum convolutional networks can provably avoid them [14]. A recent review consolidates these results [6]. This body of work is overwhelmingly analytical: it derives variance scaling laws for particular ansatz families. Our work is complementary and empirical; rather than proving a bound for one family, we learn a predictive map across a heterogeneous space of architectures.

C. THE PARAMETER-SHIFT RULE

For a gate generated by an operator with two distinct eigenvalues, the exact gradient of the cost is obtained from two circuit evaluations [15], [16]:

$$\frac{\partial C}{\partial \theta_k} = \frac{1}{2} [C(\theta + \frac{\pi}{2} e_k) - C(\theta - \frac{\pi}{2} e_k)]. \quad (2)$$

We use this rule to obtain exact per-sample gradients when estimating the variance label, so our labels are free of finite-difference bias.

D. TRAINABILITY, SIMULABILITY, AND HONEST FRAMING

Cerezo *et al.* [17] argue that the very structure guaranteeing absence of barren plateaus often also renders a model classically simulable. Bowles *et al.* [12] show that out-of-the-box classical models often match or beat QML models on standard tasks, and Schuld and Killoran [11] argue the field should de-emphasize “beating classical.” We adopt this stance: the object we predict is a property of quantum circuits, but the predictor and its value proposition are entirely classical. We are not aware of prior work that casts barren-plateau severity as a directly predicted regression/classification target from cheap architectural features and tests extrapolation across qubit count.

III. MATERIALS AND METHODS

Figure 1 summarizes the full pipeline, from sampling a circuit spec to predicting its trainability.

A. CIRCUIT SPECIFICATION AND FEATURE SPACE

Each datapoint is a *spec* that fully determines a circuit family. The feature vector handed to the model is listed in Table 1. All features are computed in closed form from the spec; no quantum simulation is needed to obtain them.

B. LABEL GENERATION

For a spec, we draw 200 parameter vectors $\theta \sim \text{Unif}[0, 2\pi)^{nL}$, compute the exact parameter-shift gradient of the cost with respect to a fixed target rotation in the middle layer, and record the empirical variance. The regression label is $y = \log_{10} \max(\text{Var}[\partial C / \partial \theta], 10^{-18})$; the classification label is 1 when $y < -2$ (“barren”). Statevectors are evolved exactly with the in-house `qsim` simulator [18], so labels reflect ideal (noiseless) trainability.

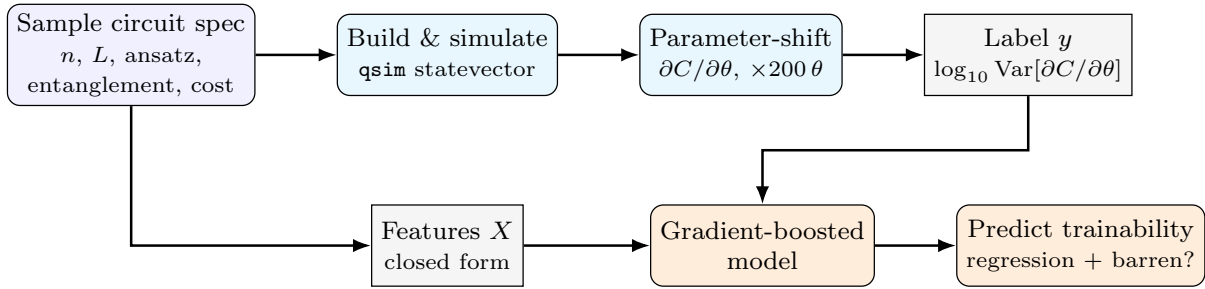


FIGURE 1. Method pipeline. A random circuit spec is built and simulated exactly with `qsim`; the parameter-shift gradient of the cost is sampled over 200 random parameter vectors to form the gradient-variance label y . Cheap closed-form architectural features X are read directly from the spec. A gradient-boosted model is trained on (X, y) to predict trainability.

TABLE 1. Architectural features used to predict trainability.

Feature	Meaning
<code>n_qubits</code>	number of qubits n
<code>n_layers</code>	number of ansatz layers L
<code>n_params</code>	$n \times L$ trainable rotations
<code>ansatz_type</code>	fixed-ry vs. random-Pauli axes
<code>entangle_pattern</code>	linear / circular / all-to-all
<code>entangler_gate</code>	CZ vs. CX
<code>cost_global</code>	local Z_0 vs. global $Z^{\otimes n}$
<code>n_entanglers</code>	entangling gates $\times$ layers
<code>depth_ratio</code>	L/n

TABLE 2. Sampled design space. Each spec draws these parameters independently and uniformly.

Design parameter	Type	Sampled values
<code>n_qubits</code>	integer	2–12
<code>n_layers</code>	integer	1–20
<code>ansatz_type</code>	categorical (2)	fixed-ry, random-Pauli
<code>entangle_pattern</code>	categorical (3)	linear, circular, all-to-all
<code>entangler_gate</code>	categorical (2)	CZ, CX
<code>cost_global</code>	binary	local Z_0 , global $Z^{\otimes n}$

C. DATASET AND DATA DESCRIPTION

We sample 20,000 specs by drawing each design parameter independently and uniformly over the ranges in Table 2, with $n \in [2, 12]$ qubits and $L \in [1, 20]$ layers, each labeled with 200 gradient samples. Generation is embarrassingly parallel; every spec is seeded reproducibly from a root seed sequence, and rows are streamed to disk as they complete. The resulting dataset has 46.7% barren circuits and 50.1% global-cost circuits, a near-balanced classification target that avoids a degenerate majority baseline.

Table 3 summarizes the numeric features and the regression label. The design deliberately spans small, easily simulated circuits ($n = 2$) up to the $n = 12$ boundary of exact statevector simulation, and depths from a single layer to $L = 20$. The label ranges over roughly eighteen orders of magnitude, with a floor at -18 where finite-sample variance underflows.

Crucially, the labels reproduce established barren-plateau theory directly from the data (Table 4). Global cost observables yield a mean log-variance more than two orders of mag-

TABLE 3. Descriptive statistics of the numeric features and the trainability label over all 20,000 circuits.

Variable	Mean	Std	Min	25%	Median	Max
<code>n_qubits</code>	7.02	3.17	2.00	4.00	7.00	12.00
<code>n_layers</code>	10.48	5.80	1.00	5.00	11.00	20.00
<code>n_params</code>	73.56	55.55	2.00	28.00	60.00	240.00
<code>n_entanglers</code>	135.82	198.34	1.00	28.00	70.00	1320.00
<code>depth_ratio</code>	2.00	1.83	0.08	0.75	1.50	10.00
<code>log10 Var (label)</code>	-3.41	4.84	-18.00	-2.98	-1.85	-0.23

TABLE 4. Trainability broken down by design choice. Lower mean log-variance and higher barren percentage indicate less trainable families.

Design choice	Count	Mean $\log_{10}$ Var	Barren %
<i>Cost-observable locality</i>			
local Z_0	9974	-2.33	30.9
global $Z^{\otimes n}$	10026	-4.49	62.4
<i>Entanglement pattern</i>			
linear	6649	-2.50	38.5
circular	6748	-2.96	48.8
all-to-all	6603	-4.79	52.8
<i>Entangler gate</i>			
CZ	10096	-2.31	38.1
CX	9904	-4.53	55.5
<i>Single-qubit rotation scheme</i>			
fixed-ry	9878	-2.80	42.0
random-Pauli	10122	-4.01	51.3

nitude smaller than local costs (-4.49 vs. -2.33) and nearly double the barren fraction, exactly the cost-locality effect proved by Cerezo *et al.* [7]. Denser entanglement and the CX entangler likewise push circuits toward barren plateaus. This structure is what makes the prediction task learnable and gives the trained model an interpretable, physics-consistent basis.

D. MODELS AND EVALUATION

To avoid privileging a single learner, we compare six model families spanning linear, instance-based, neural, and tree-ensemble inductive biases: a linear baseline (ridge regression / logistic regression), k -nearest neighbours, a multi-layer perceptron (two hidden layers, 128–64), random forests, extremely randomized trees, and histogram-based gradient-

TABLE 5. Model comparison across both tasks and both evaluation splits. Best hold-out result in bold.

Model	Regression R^2		Classification AUC	
	Hold-out	Extrap.	Hold-out	Extrap.
Hist Gradient Boosting	0.685	0.719	0.991	0.989
MLP	0.665	0.752	0.990	0.991
k-NN	0.645	0.639	0.986	0.979
Random Forest	0.640	0.716	0.985	0.988
Extra Trees	0.610	0.707	0.968	0.988
Linear baseline	0.274	0.297	0.953	0.889

boosted trees (scikit-learn’s implementation [19], inspired by LightGBM [20]). Features are standardized for the scale-sensitive models. We report two protocols for each: (i) a random hold-out split (20% test) measuring generalization to unseen architectures, and (ii) an extrapolation split, training on $n \leq 10$ and testing on all larger circuits. All models use fixed random seeds so the reported numbers reproduce exactly. Interpretability is assessed with permutation feature importance on the best model.

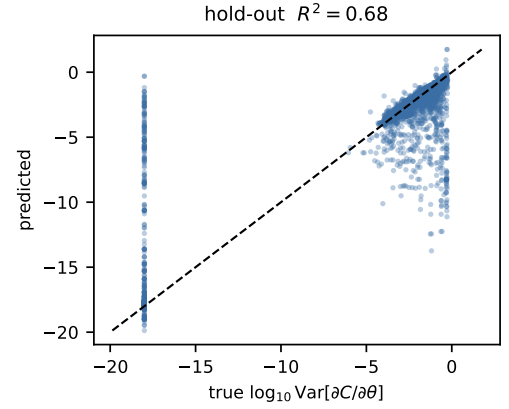
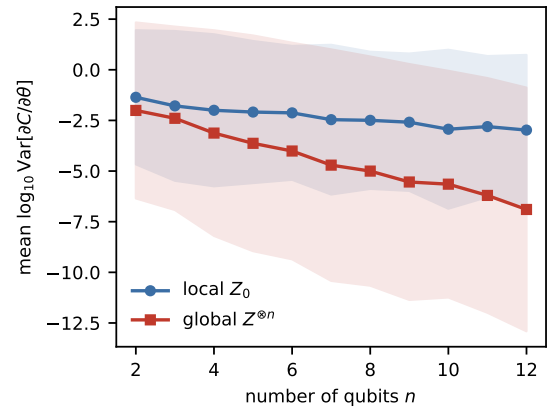
IV. RESULTS

A. MODEL COMPARISON

Table 5 compares all six model families on both tasks and both splits. Histogram-based gradient-boosted trees give the best held-out performance on both regression ($R^2 = 0.685$) and classification (AUC = 0.991), so we adopt it for the remainder of the paper. Two observations sharpen the picture. First, the linear baseline is far behind on regression ($R^2 = 0.27$ vs. 0.685); classification is easy for every model (AUC ≥ 0.95), so the architecture-to-trainability map is nonlinear mainly in its fine-grained (regression) structure. Second, the neural network is competitive and in fact edges out boosting on the extrapolation regression ($R^2 = 0.75$ vs. 0.72), hinting that a learned representation transfers slightly better to larger circuits; we retain gradient boosting for its stronger in-distribution accuracy, interpretability, and training speed.

B. GENERALIZATION AND EXTRAPOLATION

On the random hold-out split, the regressor attains $R^2 = 0.685$ with mean absolute error 1.018 in log-ten units, and the classifier separates barren from trainable circuits with accuracy 0.950 and AUC 0.991. Figure 2 shows predicted versus true log-variance on the held-out set. Training only on $n \leq 10$ qubits (16 395 circuits) and testing on the two largest unseen sizes ($n = 11, 12$; 3 605 circuits), the regressor retains $R^2 = 0.719$ (MAE 1.264). The extrapolation R^2 slightly exceeds the hold-out value; we do not read this as an easier task in general, since the $n = 11, 12$ test set is dominated by saturated, strongly barren global-cost circuits whose labels are comparatively easy to predict. It should be read as evidence that small-circuit labels transfer to the largest sizes we can still simulate. Because the ground-truth cost of labeling grows exponentially with qubit count, even this limited downward-in-cost transfer is what an ansatz-

**FIGURE 2. Predicted vs. true $\log_{10} \text{Var}[\partial C / \partial \theta]$ on held-out architectures. The dashed line is $y = x$.****FIGURE 3. Mean $\log_{10} \text{Var}[\partial C / \partial \theta]$ vs. qubit count for local and global cost observables (shaded: ± 1 standard deviation). Global costs show the exponential decay characteristic of barren plateaus.**

screening tool needs.

C. TRAINABILITY VERSUS QUBIT COUNT

Figure 3 shows the mean label as a function of qubit count, split by cost locality. Global-cost circuits exhibit the steep, monotone decay of gradient variance with qubit number that is the defining signature of a barren plateau, while local-cost circuits decay far more slowly. The dataset thus captures the qubit-scaling behavior the predictor must learn to extrapolate.

D. WHAT DRIVES TRAINABILITY

Permutation feature importance (Figure 4) ranks cost-observable locality and entanglement structure at the top, consistent with theory: global costs [7] and denser entanglement [8] drive circuits toward barren plateaus. The model rediscovers, from data alone, the design rules the barren-plateau literature derived analytically.

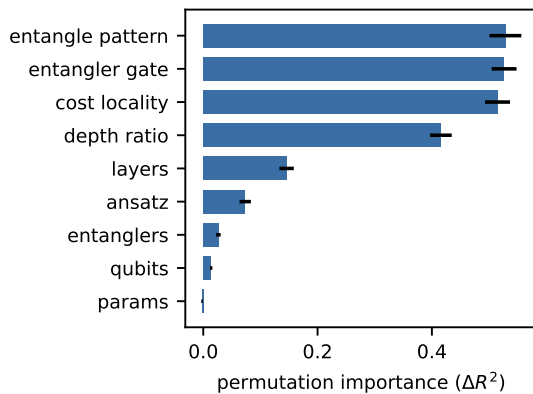


FIGURE 4. Permutation feature importance for the trainability regressor. Cost locality and entanglement structure dominate.

V. DISCUSSION

The predictor's accuracy and its physics-consistent feature ranking indicate that barren-plateau severity carries a strong, low-dimensional architectural signature, enough for a cheap classical model to recover without any gradient sampling. This suggests a practical workflow: screen a large space of candidate ansätze with the learned model, and reserve expensive gradient-variance estimation for the shortlist it flags as trainable.

Several limitations scope the claims. First, labels are limited to classically simulable qubit counts, so the extrapolation result is evidence of transfer, not a proof. Second, we use an ideal statevector simulator, so noise-induced plateaus [10] are out of scope. Third, the spec space, while diverse, is a hardware-efficient family; other ansatz classes may need their own features. Fourth, gradient variance is estimated from finitely many samples, adding label noise at the trainable end. Fifth, the label is the variance of the gradient with respect to a single fixed middle-layer rotation, and in principle the variance can depend on which parameter is probed. Sixth, the barren/trainable cutoff ($\log_{10} \text{Var} < -2$) is a modeling choice; our primary regression target is threshold-free, so the cutoff affects only the binary framing. Seeds are fixed and reported, and the extrapolation split is defined purely by qubit count, so no large-circuit information leaks into training.

VI. CONCLUSION

We recast barren-plateau diagnosis as supervised prediction and showed that a classical gradient-boosted model, trained on statevector-simulated labels, can predict a variational circuit's trainability from its architecture alone, and extrapolate across qubit count. The approach offers a cheap ansatz-screening heuristic and a data-driven confirmation of which design choices cause untrainability. Future work includes richer ansatz families, graph or sequence circuit encodings with graph-neural-network or transformer models, and incorporating a noise model to predict noise-induced plateaus.

REPRODUCIBILITY

All code and scripts are openly available at https://github.com/bithabib/quantum_computer_simulator. All code (statevector simulator, dataset generator, training and comparison scripts), the exact command lines, and random seeds are released with this article. The dataset is regenerable end-to-end with a single command, and interactive demonstrations of the underlying simulator are available at <https://quantum.qbithabib.com>.

REFERENCES

- [1] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, pp. 625–644, 2021.
- [2] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, no. 7747, pp. 209–212, 2019.
- [3] E. Farhi and H. Neven, "Classification with Quantum Neural Networks on Near Term Processors," *arXiv preprint arXiv:1802.06002*, Feb. 2018.
- [4] M. Schuld, A. Bocharov, K. M. Svore, and N. Wiebe, "Circuit-centric quantum classifiers," *Physical Review A*, vol. 101, no. 3, p. 032308, 2020.
- [5] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nature Communications*, vol. 9, no. 1, p. 4812, 2018.
- [6] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, and M. Cerezo, "Barren plateaus in variational quantum computing," *Nature Reviews Physics*, vol. 7, pp. 174–189, 2025, arXiv:2405.00781.
- [7] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, "Cost function dependent barren plateaus in shallow parametrized quantum circuits," *Nature Communications*, vol. 12, no. 1, p. 1791, 2021.
- [8] Z. Holmes, K. Sharma, M. Cerezo, and P. J. Coles, "Connecting ansatz expressibility to gradient magnitudes and barren plateaus," *PRX Quantum*, vol. 3, no. 1, p. 010313, 2022.
- [9] E. Grant, L. Wossnig, M. Ostaszewski, and M. Benedetti, "An initialization strategy for addressing barren plateaus in parametrized quantum circuits," *Quantum*, vol. 3, p. 214, 2019.
- [10] S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, and P. J. Coles, "Noise-induced barren plateaus in variational quantum algorithms," *Nature Communications*, vol. 12, no. 1, p. 6961, 2021.
- [11] M. Schuld and N. Killoran, "Is quantum advantage the right goal for quantum machine learning?" *PRX Quantum*, vol. 3, no. 3, p. 030101, 2022.
- [12] J. Bowles, S. Ahmed, and M. Schuld, "Better than classical? The subtle art of benchmarking quantum machine learning models," *arXiv preprint arXiv:2403.07059*, Mar. 2024.
- [13] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, and J. M. Gambetta, "Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets," *Nature*, vol. 549, no. 7671, pp. 242–246, 2017.
- [14] A. Pesah, M. Cerezo, S. Wang, T. Volkoff, A. T. Sornborger, and P. J. Coles, "Absence of barren plateaus in quantum convolutional neural networks," *Physical Review X*, vol. 11, no. 4, p. 041011, 2021.
- [15] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, "Quantum circuit learning," *Physical Review A*, vol. 98, no. 3, p. 032309, 2018.
- [16] M. Schuld, V. Bergholm, C. Gogolin, J. Izaac, and N. Killoran, "Evaluating analytic gradients on quantum hardware," *Physical Review A*, vol. 99, no. 3, p. 032331, 2019.
- [17] M. Cerezo, M. Larocca, D. García-Martín *et al.*, "Does provable absence of barren plateaus imply classical simulability?" *Nature Communications*, vol. 16, p. 7907, 2025, arXiv:2312.09121.
- [18] Eshyana Project, "Eshyana: An educational quantum computing simulator," 2026, <https://quantum.qbithabib.com>.
- [19] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel *et al.*, "Scikit-learn: Machine learning in python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [20] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "Lightgbm: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

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